

ELITE MATHEMATICS

Pearson Edexcel International A Level

Pure 1

Adaptive Classified

Chapter 07 · Transformations of Functions

Questions

Student book · questions with writing space

- 22 pages
- Past papers up to June 2026

Transformations of Functions

Your classified route

7
questions

55
marks

3
families

6
variants

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CLASSIFIED BY IDEA

Move from recognition to method, then use the working space to complete the solution.

Transformations of Functions

Families and variants

F01 • Coordinate maps under function notation

V01 Translations and reflections of key points and asymptotes

V02 Horizontal input scale/reflection with equations and inequalities

V03 Combined input transformation and intercept/multiplicity mapping

F02 • Transformed algebraic graphs and exact intersections

V01 Square-root transformations with exact surd intersection

V02 Translated reciprocal graph followed by line intersection

F03 • Transformations on trigonometric graphs

V01 Identify a sine graph and transform a labelled extremum

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Coordinate maps under function notation

WHAT THIS FAMILY ASKS

Transform an already-known graph and carry its labelled features through the coordinate map.

VARIANT ROUTE

V01 Translations and reflections of key points and asymptotes

V02 Horizontal input scale/reflection with equations and inequalities

V03 Combined input transformation and intercept/multiplicity mapping

3 Complete Classified questions

20 total marks



The curve has a minimum at $P(-1, 0)$ and a maximum at $Q\left(\frac{3}{2}, 2\right)$

On separate diagrams sketch the curves with equation

(i) $y = f(x) - 2$

(3)



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(3)

A graph of a function $y = f(x)$ is shown on a Cartesian coordinate system. The function is a continuous curve that passes through the points $(-4, 6)$, $(-1, 6)$, $(2, 6)$, and $(3, 6)$. The curve has a local maximum between $x = -4$ and $x = -1$, a local minimum between $x = -1$ and $x = 2$, and another local maximum between $x = 2$ and $x = 3$. The origin is labeled O .

Figure 1 shows the curve with equation $y = f(x)$.

(a) Using Figure 1, find the range of values of x for which

$$f(x) < 6$$

(3)



The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

$$f(2x) = 6$$

(1)



The points $P(-4, 6)$, $Q(-1, 6)$, $R(2, 6)$ and $S(3, 6)$ lie on the curve.

- On your sketch, state the coordinates of the points to which P , Q , R and S are transformed.

- $$f(-x) \geq 6 \text{ and } x < 0$$

(4)

Figure 1 shows a sketch of the curve with equation $y = f(x)$.

On separate diagrams, sketch the curve with equation

(3)

Figure 1 shows a sketch of the curve with equation $y = f(x)$.

On separate diagrams, sketch the curve with equation

(3)

Transformed algebraic graphs and exact intersections

WHAT THIS FAMILY ASKS

Describe the graph transformations of a named algebraic function and then solve exactly where two transformed versions meet.

VARIANT ROUTE

V01 Square-root transformations with exact surd intersection

V02 Translated reciprocal graph followed by line intersection

3 Complete Classified questions

29 total marks

$$f(x) = \sqrt{x} \quad x > 0$$

The point $P(9, 3)$ lies on the curve and is shown in Figure 5.

(a) On Diagram 1, sketch and clearly label the graphs of

$$y = f(2x) \quad \text{and} \quad y = f(x) + 3$$

Show on each graph the coordinates of the point to which P is transformed.

(3)

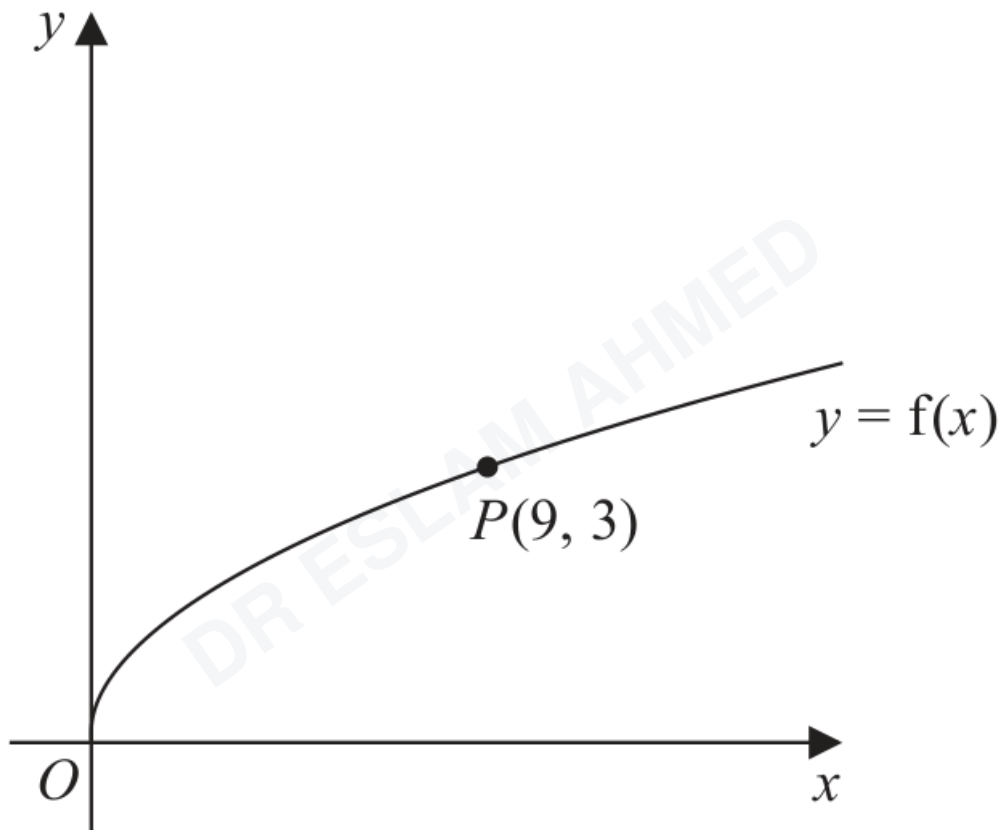


Diagram 1

$$f(x) = \sqrt{x} \quad x > 0$$

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q .

(b) Show that the x coordinate of Q is the solution of

$$\sqrt{x} = 3(\sqrt{2} + 1)$$

(3)

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$$f(x) = \sqrt{x} \quad x > 0$$

The graph of $y = f(2x)$ meets the graph of $y = f(x) + 3$ at the point Q .

$$\sqrt{x} = 3(\sqrt{2} + 1)$$

(c) Hence find, in simplest form, the coordinates of Q .

(3)

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Figure 4 shows a sketch of

- (a) Describe fully the single transformation that would transform

- (4)

- the graph C_1 with equation $y = \sqrt{2x}$
- the graph C_2 with equation $y = 12 - \sqrt{x}$

The graphs C_1 and C_2 meet at the point P , as shown in Figure 4.

(b) (i) Show that the x coordinate of P is a solution of

$$\sqrt{x} = 12(\sqrt{2} - 1)$$

(ii) Hence find, in simplest form, the exact coordinates of P .

(6)

7. (a) On Diagram 1, sketch a graph of the curve C with equation

$$y = \frac{6}{x} \quad x \neq 0$$

(2)

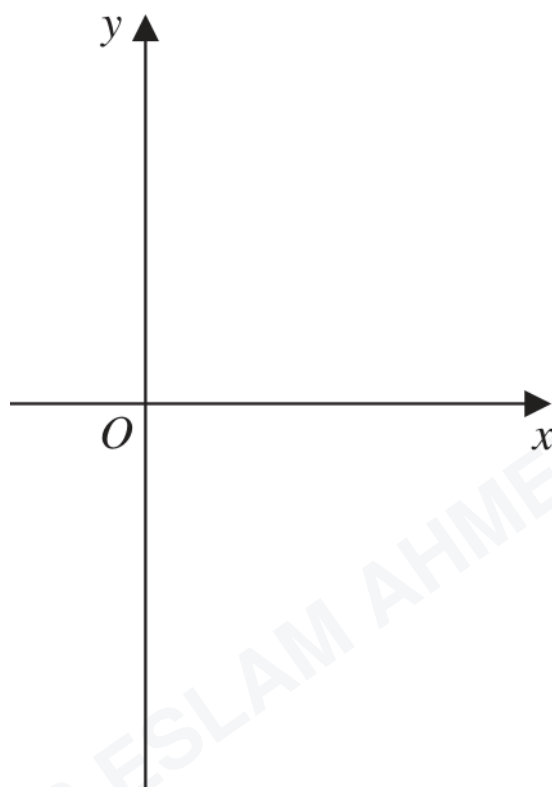


Diagram 1

$$y = \frac{6}{x} \quad x \neq 0$$

The curve C is transformed onto the curve with equation $y = \frac{6}{x-2} \quad x \neq 2$

(b) Fully describe this transformation.

(2)

The curve with equation

$$y = \frac{6}{x-2} \quad x \neq 2$$

and the line with equation

$$y = kx + 7 \quad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q .

Given that the x coordinate of point P is -4

(c) find the value of k ,

(2)

The curve with equation

$$y = \frac{6}{x-2} \quad x \neq 2$$

and the line with equation

$$y = kx + 7 \quad \text{where } k \text{ is a constant}$$

intersect at exactly two points, P and Q .

Given that the x coordinate of point P is -4

(d) find, using algebra, the coordinates of point Q .

(Solutions relying entirely on calculator technology are not acceptable.)

(4)

Transformations on trigonometric graphs

WHAT THIS FAMILY ASKS

Recognise a basic sinusoidal graph and map a labelled point under phase translation and output reflection/scaling.

VARIANT ROUTE

V01 Identify a sine graph and transform a labelled extremum

1 Complete Classified questions

6 total marks



(a) Write down an expression for $f(x)$

(2)



The point P lies on $y = f(x)$ and is shown in Figure 3.

$$(i) \quad y = f\left(x - \frac{\pi}{6}\right) \quad (2)$$

$$\text{(ii) } y = -\frac{1}{2}f(x) \quad (2)$$

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